

Biosimilars approved for hematology use: lessons from comparative efficacy studies and comparative analytical assessments

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Biosimilars used in hematology have been at the forefront of biosimilar development since the first US approval in 2015 of a filgrastim product, with >80 now licensed by the US Food and Drug Administration (FDA). Developers traditionally submitted data from a comparative analytical assessment (CAA) and clinical studies, often including a comparative efficacy study (CES), to support demonstrating their product is “highly similar” to and has “no clinically meaningful differences” from its reference product (RP). However, growing scientific confidence in the analytical comparisons between biosimilars and their RP included in the CAA has prompted the FDA and global

regulators to reconsider the utility of CESs. Recently, the FDA published updated recommendations on the need for a CES in biosimilar programs. As approvals for hematologic biosimilars span the entire history of FDA biosimilar development across a breadth of patient ages and diseases, reexamining these programs provides valuable insights into the evolving role of the CES in the past and its role in the future. In this review, we present a historical overview of the development programs for all FDA-approved hematologic biosimilars in the context of an emerging global consensus recognizing that the CAA is more sensitive than the CES for predicting biosimilarity.

Introduction

Biosimilars are biologic medications that are “highly similar” to an approved biologic reference product (RP) with “no clinically meaningful differences” in terms of safety, purity, or potency. The Biologics Price Competition and Innovation Act of 2009 (BPCIA) created an abbreviated pathway for the development of biosimilar products in the United States.¹ Regulatory pathways for approving biosimilars were established in the European Union in 2005, Japan in 2009, and Canada in 2010. In 2009, the World Health Organization published guidelines for the evaluation of biosimilars that are the basis for approvals for many jurisdictions worldwide. For most jurisdictions, the RP for a biosimilar is an originator biologic whose approval was based on data primarily from extensive clinical trials. Biosimilars increase access to expensive biologics by providing market competition and offering the same risks and benefits as their originator biologics at potentially lower cost.¹ Since the approval of the first biosimilar in the United States, cumulative savings have reached \$56 billion with biosimilars costing up to 35% less than their RP.²⁻⁴ Globally, cumulative savings are expected to reach \$383 billion by 2027.⁵ Despite an increase in the approval and adoption of biosimilars, barriers such as patent disputes continue to delay the marketing of approved biosimilars.⁶

The first biosimilar approved by the US Food and Drug Administration (FDA) in March 2015 was a biosimilar to

filgrastim (filgrastim-sndz), a recombinant granulocyte-colony stimulating factor used for treating neutropenia and mobilization of hematopoietic progenitor cells.⁶⁻⁹ Currently, 85 biosimilars to 20 different RPs are therapeutic proteins approved by the FDA, with 33 biosimilars corresponding to 7 different RPs approved for indications treated by hematologists⁹ (Table 1). The biosimilars licensed for hematologic conditions are growth factors (filgrastim, epoetin alfa, pegfilgrastim) for treatment of children and adult patients with neutropenia and anemia, and monoclonal antibodies (rituximab, tocilizumab, denosumab, eculizumab) for treatment of patients with complex hematologic diseases, including non-Hodgkin lymphoma, chronic lymphocytic leukemia, multiple myeloma, and paroxysmal nocturnal hemoglobinuria (PNH). Most hematologists are familiar with biosimilars and consider them safe and appropriate to use for patients naïve to treatment or actively receiving treatment with the RP, with efficacy expected to be equivalent for a biosimilar and the corresponding RP.¹⁰

Early FDA guidance advised biosimilar developers to use a “stepwise approach” for data expected to support biosimilarity¹¹ and for addressing the “highly similar” and “no clinically meaningful differences” standards described in the BPCIA. A comparative analytical assessment (CAA) consisting of extensive comparisons of structural and functional attributes between the proposed biosimilar and the RP and other closely related product quality assessments were the foundation of

Table 1. FDA-approved biosimilars with hematology indications listed chronologically within RP class

RP and biosimilar	First approval date (month/year)	RP labeled hematologic conditions
Filgrastim Filgrastim-sndz Filgrastim-aafi Filgrastim-ayow Filgrastim-txid Filgrastim-laha	02/1991 03/2015 07/2018 02/2022 06/2024 01/2026	Patients with cancer receiving myelosuppressive chemotherapy Patients with acute myeloid leukemia receiving induction or consolidation chemotherapy Patients with cancer undergoing bone marrow transplant Patients undergoing autologous peripheral blood progenitor cell collection and therapy Patients with severe chronic neutropenia Patients acutely exposed to myelosuppressive doses of radiation
Epoetin alfa Epoetin alfa-epbx	06/1989 05/2018	Anemia due to chronic kidney disease Anemia due to zidovudine in patients with HIV infection Anemia due to chemotherapy in patients with cancer Reduction of allogeneic red blood cell transfusions in patients undergoing elective, noncardiac, nonvascular surgery
Pegfilgrastim Pegfilgrastim-jmdb Pegfilgrastim-cbqv Pegfilgrastim-bmez Pegfilgrastim-apgf Pegfilgrastim-pbbk Pegfilgrastim-fpgk Pegfilgrastim-unne	01/2002 06/2018 11/2018 11/2019 06/2020 05/2022 09/2022 11/2025	Patients with cancer receiving myelosuppressive chemotherapy Patients with hematopoietic subsyndrome of acute radiation syndrome
Rituximab Rituximab-abbs Rituximab-pvvr Rituximab-arxx	11/1997 11/2018 07/2019 12/2020	NHL Adult patients with: Relapsed or refractory, low-grade or follicular, CD20 ⁺ , B-cell NHL as a single agent Previously untreated follicular, CD20 ⁺ , B-cell NHL in combination with first-line chemotherapy and, in patients achieving a complete or partial response to a rituximab product in combination with chemotherapy, as single-agent maintenance therapy Nonprogressing, low-grade, CD20 ⁺ , B-cell NHL as a single agent after first-line cyclophosphamide, vincristine, and prednisone chemotherapy Previously untreated diffuse large B-cell, CD20 ⁺ NHL in combination with cyclophosphamide, doxorubicin, vincristine, prednisone, or other anthracycline-based chemotherapy regimens Pediatric patients aged ≥6 months with: Previously untreated, advanced stage, CD20 ⁺ diffuse large B-cell lymphoma, Burkitt lymphoma, Burkitt-like lymphoma or mature B-cell acute leukemia in combination with chemotherapy Chronic lymphocytic leukemia
Tocilizumab Tocilizumab-bavi Tocilizumab-aazg Tocilizumab-anoh	01/2010 09/2023 03/2024 01/2025	Chimeric antigen receptor T-cell–induced severe or life-threatening cytokine release syndrome
Denosumab Denosumab-bbdz Denosumab-dssb Denosumab-bmwo Denosumab-bnht Denosumab-nxxp Denosumab-kyqq Denosumab-qbde Denosumab-desu Denosumab-mobz	06/2010 03/2024 02/2025 02/2024 03/2025 08/2025 09/2025 09/2025 10/2025 12/2025	Skeletal-related events in patients with multiple myeloma and in patients with bone metastases from solid tumors Hypercalcemia of malignancy refractory to bisphosphonate therapy
Eculizumab Eculizumab-aeeb Eculizumab-aagh	03/2007 05/2024 07/2024	PNH to reduce hemolysis Atypical hemolytic uremic syndrome to inhibit complement-mediated thrombotic microangiopathy

Biosimilars included if approved before 1 March 2026.

Biosimilars listed may not be approved for all the indications approved for the RP.

See Drugs@FDA (www.fda.gov/drugsatfda) for current information on approved indications for RPs and biosimilars.

NHL, non-Hodgkin lymphoma.

this stepwise approach. Subsequent steps in biosimilar development included animal toxicity studies, human pharmacokinetic (PK) studies with a clinical immunogenicity assessment comparing the incidence of antidrug antibodies, and a

comparative efficacy study (CES). FDA guidance encouraged developers to seek FDA advice after completing the CAA and before finalizing their clinical program, but the time required to complete a CES and the goal of obtaining the earliest date for

a developer to submit a biologics license application often resulted in initiating the CES early in development before completion of the CAA.

The relative importance of CES and CAA in biosimilar development and approval has been a source of debate among developers, regulators, and especially clinicians. Clinicians are used to large, randomized Phase 3 studies for definitive scientific evidence of the efficacy and safety of a biologic and are cautious and skeptical when statistically powered efficacy data are not available.^{12,13} With biosimilar development, approval, and postmarketing experience, a consensus has emerged that CESs generally lack the sensitivity to discriminate between products, even when there are known differences in attributes assessed by the CAA. The analytical methods used in the CAA can identify minor variations and other microheterogeneity expected in biotechnology-derived therapeutic proteins. Therefore, the CAA can detect small differences between a biosimilar and its RP and identify differences that affect clinical performance. This is the basis for the FDA's recently published draft guidance¹⁴ on assessing the need for CES. This process of accepting the relative merits of CESs and CAAs may be viewed as a Hegelian or dialectic process, after the work of Hegel.¹⁵ When a belief or thesis is confronted with an opposing view or antithesis, following a period of discord, a higher-level understanding and synthesis is eventually achieved. This review using biosimilars with hematologic indications takes a Hegelian approach to recount the changing relevance of the CES (thesis) and the rising acceptance of the CAA (antithesis) to best identify differences between a biosimilar and RP and arrive at the current synthesis where CESs are not as sensitive as CAAs in predicting differences in performance, do not inform approvability, and for most programs a CAA and a human PK study are sufficient for determining biosimilarity.¹⁶⁻²¹

Thesis: CESs resolve uncertainty in determining biosimilarity

The CES in biosimilar development was originally intended to resolve any uncertainty about biosimilarity following completion of a PK similarity study.^{11,14} Healthy subjects are the preferred population for the PK study; if safety is a concern, the study is conducted in patients (supplemental Tables 1 and 2, available on the *Blood* website). A PK study in healthy subjects is considered more sensitive in evaluating product similarity because it is likely to produce less PK variability compared with a study in patients with potential confounding factors such as underlying disease and concomitant medications.²² Clinical pharmacology studies with PK or pharmacodynamics (PD) end points generally use an equivalence approach where the 90% confidence interval of the ratio of the geometric means falls within an acceptable limit, usually 80% to 125%.²² In addition to standard PK end points, the PK study usually includes an assessment of immunogenicity consisting of a comparison of adverse events related to hypersensitivity and collection of antidrug and neutralizing antibody data and their effect on PK.

The clinical programs for biosimilars to filgrastim and pegfilgrastim, conducted almost entirely in healthy subjects, included a single-dose PK-PD crossover study and a multidose study to collect antidrug and neutralizing antibody data

(Figure 1; supplemental Tables 1 and 2). Although not required by the FDA, some applicants submitted additional studies for these programs. These included PD biomarker end point studies that used statistical analyses based on asymmetric margins rather than the standard equivalence approach used for PK-PD studies. The program for the biosimilar to epoetin alfa included 2 PK studies in healthy subjects and 2 CESs using different routes of administration in patients with chronic kidney disease and anemia who were on dialysis (Figure 1; supplemental Tables 1 and 2). Coprimary end points for both CESs were the difference between the RP and biosimilar arms in mean weekly hemoglobin and mean weekly dose during the last 4 weeks of the treatment period. Immunogenicity was assessed in all 4 clinical studies. During the development of the first biosimilar to epoetin alfa, concerns remained regarding erythropoietin-related pure red cell aplasia²³ and may have influenced conducting multiple CESs for this program.

Concurrent with the development and FDA approvals of the first filgrastim, epoetin alfa, and pegfilgrastim biosimilars (March 2015-June 2018), biosimilars to larger and more complex proteins were being developed and reviewed by the FDA. Between April 2016 and December 2017, the FDA licensed biosimilars to the monoclonal antibodies infliximab, adalimumab, bevacizumab, and trastuzumab, and the fusion protein, etanercept.⁹ During this period, there was reliance on clinical studies to build confidence in biosimilars as regulatory authorities, health care providers, and patients were becoming familiar with reviewing, prescribing, and using biosimilars. The application for the first adalimumab²⁴ biosimilar licensed by the FDA consisted of a CAA, a PK similarity study in healthy subjects, a CES in patients with rheumatoid arthritis (RA), and a second CES in patients with plaque psoriasis.²⁴ The plaque psoriasis study included an extension period where patients receiving the RP either continued the RP or were switched to the biosimilar. This crossover from RP to biosimilar provided a descriptive assessment of safety intended to address concerns with switching patients on the RP to a biosimilar. Over 1000 patients participated in this biosimilar program. The development of biosimilar monoclonal antibodies for hematology indications followed this approach of conducting large clinical studies in patients. Although the lack of sensitive and reliable PD biomarkers for these programs was a contributing factor, the FDA expected the CAA, ≥ 1 PK similarity study, and ≥ 1 statistically powered CESs for these programs. Often the CES contained ≥ 1 switches between the RP and the biosimilar to further assess immunogenicity and safety risks related to switching between the RP and biosimilar.^{25,26}

The development program for biosimilars to rituximab, the first monoclonal antibody biosimilar approved for hematology indications, included a CAA with more attribute comparisons than previous CAAs for hematologic biosimilars (supplemental Table 3). Rituximab was not considered safe for administration to healthy subjects; therefore, PK studies were performed in patients with RA (Figure 1; supplemental Tables 1 and 2). The study contained a single crossover period where patients were switched from RP to biosimilar compared with patients who continued the RP. As a sensitive and reliable PD biomarker, such as absolute neutrophil count for biosimilars to filgrastim, was not unavailable for rituximab, the FDA also recommended a CES using a population with a hematologic malignancy;

Reference product	Approval date of first biosimilar in RP class	Approved biosimilars (N)	Comparative efficacy studies per program - Mean - Min - Max	Design - Biosimilar - Reference product	Population	Primary end point
Filgrastim	3/2015	5		4 arm multidose with switches 	Women with breast cancer receiving myelosuppressive chemotherapy	Duration of severe neutropenia
Epoetin alfa	5/2018	1		Study 1 and 2: 2 arm multidose 	Study 1 and 2: Chronic kidney disease	Study 1 and 2: Mean weekly hemoglobin and dose
Pegfilgrastim	6/2018	7		2 arm multidose 	Women with breast cancer receiving myelosuppressive chemotherapy	Duration of severe neutropenia
Rituximab	11/2018	3		Study 1 and 2: 2 arm multidose 	Study 1: Advanced follicular lymphoma Study 2: Low tumor burden-follicular lymphoma	Study 1 and 2: Overall response rate
Tocilizumab	9/2023	3		2 arm multidose 	Rheumatoid arthritis	ACR or DAS28-ESR
Denosumab	3/2024	9		2 arm multidose 	Women with osteoporosis	Bone mineral density
Eculizumab	5/2024	2		2 arm cross over 	Paroxysmal nocturnal hemoglobinuria	Lactate dehydrogenase

Figure 1.

efficacy results from the PK study in the RA population were supportive. The program for the first biosimilar to rituximab contained a randomized double-blind CES in patients with advanced follicular lymphoma (AFL). The sensitivity of the AFL population to detect differences between products due to possible influence by the presence of background chemotherapy and the study size required to overcome this effect resulted in a study that was difficult to interpret. A third randomized double-blind study in patients with low tumor burden-follicular lymphoma (LTB-FL)²⁷ was conducted and included in the biologics license application. Patients with previously untreated LTB-FL were considered a more sensitive population than the AFL population for distinguishing clinically meaningful differences. As rituximab was given as monotherapy in treating the LTB-FL population,²⁸ additional safety signals from concomitant chemotherapy would be minimized²⁹ and the study in patients with LTB-FL was considered more sensitive to detect potential differences than the study in patients with AFL. The major immunogenicity assessment was from patients with RA based on the reasoning that patients with lymphoma receiving chemotherapy would be more likely to have a suppressed immune response than patients with RA.

The programs for biosimilars to tocilizumab contained 1 to 2 CESs (Figure 1; supplemental Tables 1 and 2). Approvals were based on the results of the CAA (supplemental Table 3), ≥ 1 dedicated PK study, and a CES in patients with RA.

Although development programs for biosimilars to rituximab and tocilizumab continued the approach of more elaborate clinical studies, several biosimilar development programs were shifting to an approach of fewer clinical studies. This switch was partially due to the growing understanding of the high degree of sensitivity of the assays used to perform the CAAs and their ability to predict clinical performance.³⁰⁻³⁴ Most biosimilar programs continued to include ≥ 1 CES. Biosimilars to denosumab are approved based on a CAA (supplemental Table 3), a PK similarity study in healthy subjects, and a CES in postmenopausal women with osteoporosis (Figure 1; supplemental Tables 1 and 2).

The clinical program for biosimilars to eculizumab consisted of a PK similarity study in healthy subjects and a crossover study in patients with PNH comparing the proposed biosimilar to the RP using a PD biomarker as the primary end point for hemolysis, as measured by a time-adjusted area under the effect curve of lactate dehydrogenase for both treatment periods (Figure 1; supplemental Tables 1 and 2). As LDH area under the effect curve is correlated with hemoglobin stabilization rate,^{35,36} it was an acceptable PD biomarker for hemolysis. Due to the scarcity of patients with PNH, recruitment for a clinical study for a biosimilar was challenging.^{37,38} Given the need for a small sample size, for the comparative clinical study, a crossover study design was used with patients receiving either the biosimilar or RP during period 1 and crossing over to the other treatment during period 2.^{39,40}

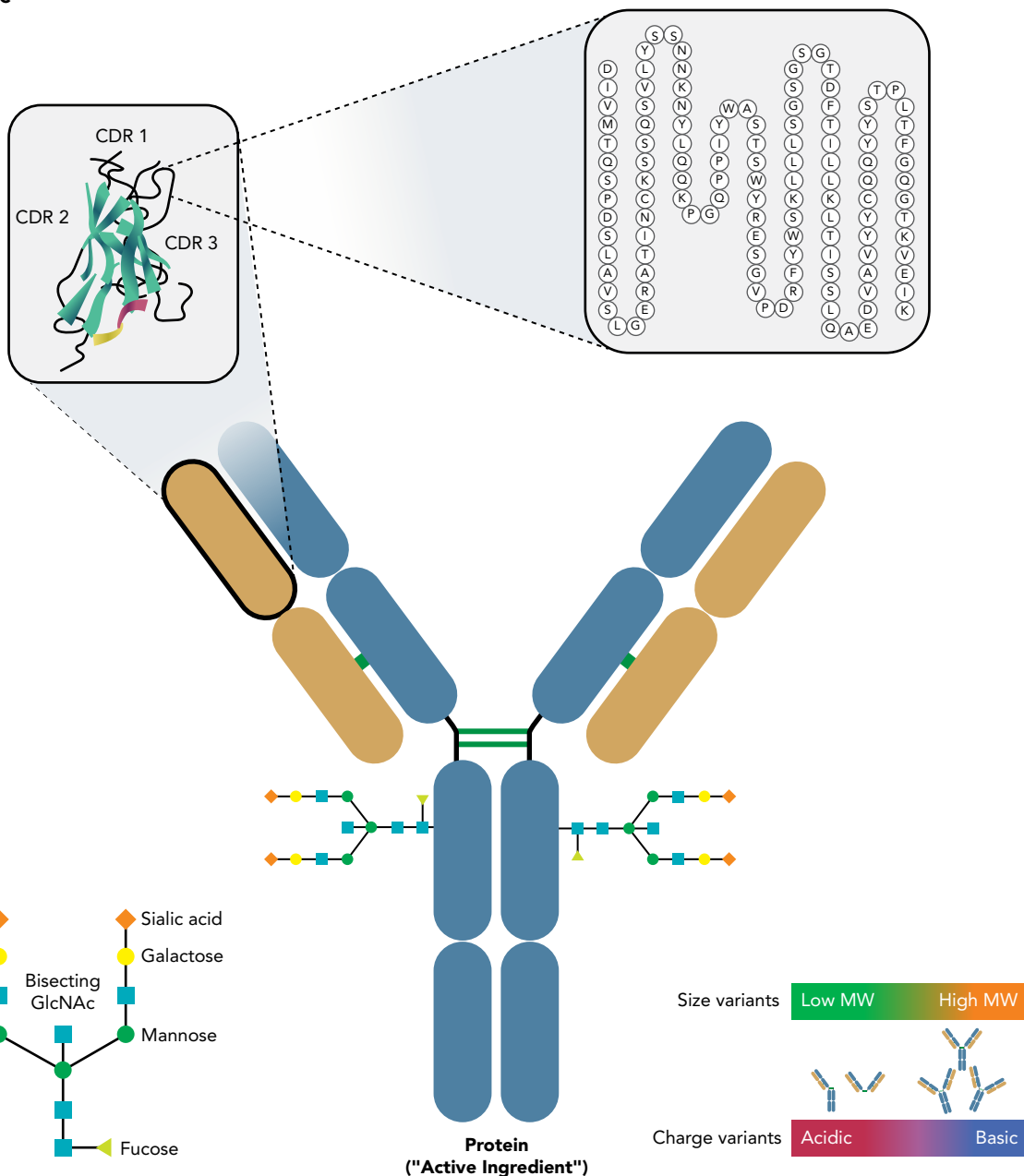
Antithesis: CAA is more sensitive than CES for determining biosimilarity

The increase in numbers and complexity of CESs submitted to the FDA was accompanied by a growing recognition of the higher sensitivity of CAAs over CESs to identify differences between a biosimilar and RP, should they exist. The principles underlying the CAA are based on the process for ensuring that measurable physical, chemical, and biologic characteristics or quality attributes of approved biologics remain within an acceptable range following a change in a manufacturing process⁴¹ and assess the impact a manufacturing change has on the clinical performance of an approved biologic.³⁰ For >3 decades, clinicians have been prescribing biologics that have had clinical performance assessed through analytical comparisons of product batches or lots made before and after a manufacturing change. Biologics that are biotechnology-derived therapeutic proteins typically consist of a heterogeneous population of minor variations of the major isoform of the protein expressed in a clonal cell line. Expected microheterogeneity within a given lot is an important distinction between biologics and small molecule drugs and underlies why biosimilars are not “generic biologics.” For small molecules and generics, the active ingredient is identical between the generic drug and its reference listed drug and between lots for each drug. For proteins manufactured in living cells, unlike synthetic generics, variability is expected between lots of the RP or biosimilar. Although based on the same principles, the analytical comparisons in the CAA are more extensive between a candidate biosimilar and its RP than to support approval of a manufacturing change.

The CAA consists of direct comparisons of the candidate biosimilar and the RP spanning multiple lots of each product. These include comparisons of primary and higher order structure, posttranslational modifications including glycosylation, and specific product-related functional activity (Figure 2). Many attributes are compared using ≥ 2 orthogonal assays that use different detection or quantification methods to accurately measure each attribute.⁴² The FDA recommends attributes that are measured quantitatively be compared using a “quality range” approach, where results for the biosimilar candidate are compared with the range of results for the RP established by the sample mean $\pm$ the sample standard deviation. Comparative analysis of a quality attribute would generally support a finding that the proposed product is highly similar to the RP when a sufficient percentage of biosimilar lot values (eg, 90%) fall within the quality range defined for that attribute.⁴³ Biosimilars, as with all biologics, are assessed for process-related impurities associated with manufacturing. Examples include impurities from cell substrates used to manufacture the proteins, cell culture components, and downstream processing steps (Figure 3).^{43,44} Recently, a systematic naming taxonomy was developed for quality attributes of therapeutic proteins,⁴⁵ which was used to categorize the quality attributes included in the CAA for hematologic biosimilars (supplemental Table 3).

Figure 1. CESs submitted for approved biosimilars used to treat hematology indications. Clinical studies with efficacy end points submitted as part of a licensing application are shown in order of first biosimilar approved for a specific RP beginning with filgrastim biosimilars. The bar graphs show the mean, minimum (min), and maximum (max) number of CESs submitted for biosimilars approved for a given RP. Additional information is provided on the study designs, populations, and primary end points (population and primary end point are taken from the first biosimilar program leading to approval with an emphasis on studies with hematology end points). ACR, American College of Rheumatology; DAS28-ESR, Disease Activity Score 28-Joint Count with Erythrocyte Sedimentation Rate.

Structure



Function

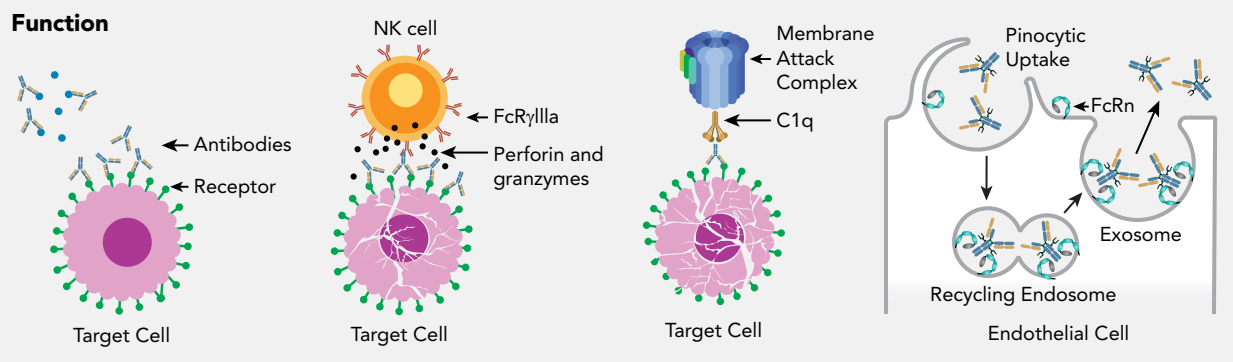
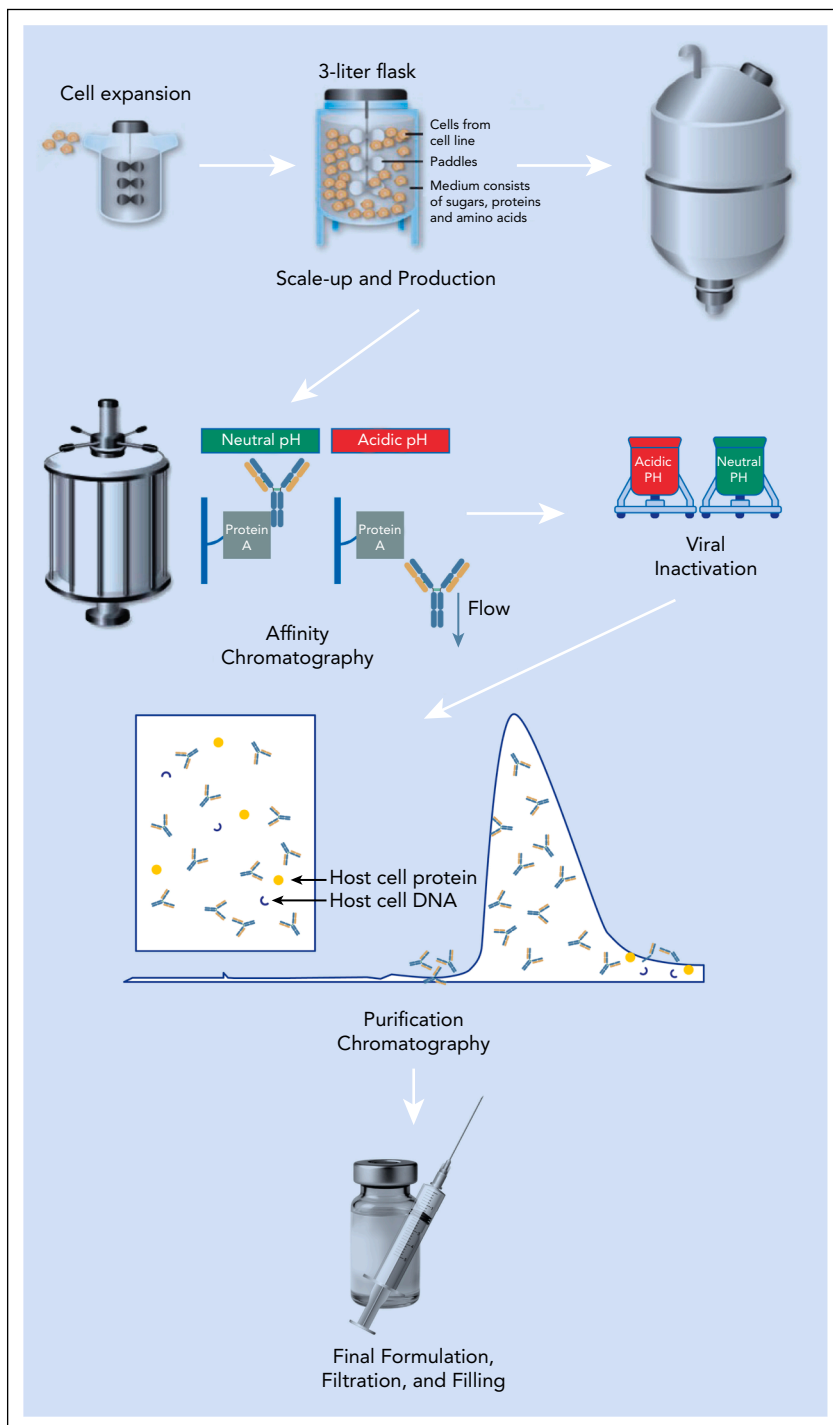


Figure 2. The components of the CAA. The CAA compares structural and functional attributes between the proposed biosimilar and its RP. The structural attributes include categories of primary and higher order structure and linked nonprotein moieties including glycosylation. The functional attributes are related to the mechanism of

Figure 3. Manufacturing processes. Process-related manufacturing steps not associated with the CAA are scale-up and production, affinity chromatography, purification, removal of process-related impurities such as host cell proteins and DNA, microbial control, and the final protein concentration and formulation filled in containers that are safe and that maintain purity and potency.



The structural attributes assessed for differences between a potential biosimilar and its RP include primary and higher order structures, glycosylation patterns, and other posttranslational modifications (Figure 2). Attributes associated with higher order structure are primarily related to disulfide bond formation and structural stability and may lead to variations in

folding, stabilization, high and low molecular weight variants between the proposed biosimilar and RP. The functional attributes compared in the CAA reflect the known and potential mechanisms of action of the proposed biosimilar and its RP. Biosimilars to growth factors are assessed for similar target binding such as granulocyte colony-stimulating factor receptor

Figure 2 (continued) action of the RP, and include evaluation of target binding, cellular activity (antibody-dependent cellular cytotoxicity, complement-dependent cytotoxicity) and binding activity to specific receptors (fragment crystallizable gamma receptor IIIa, neonatal fragment crystallizable receptor). Other quality attributes comparisons include molecular mass, and variations in size and charge. CDR, complementarity-determining region; GlcNAc, N-acetylglucosamine; MW, molecular weight; NK, natural killer.

binding for filgrastim-based biosimilars and erythropoietin receptor binding for biosimilars to epoetin alfa. Biosimilars to monoclonal antibodies are assessed for similar target binding activity, a function of the antibody's Fab region, and are considered essential initial steps for the mechanisms of action of rituximab, tocilizumab, denosumab, and eculizumab. Comparisons are also performed of functional aspects of the antibody's Fc region including binding to Fc receptors. Cell-based assays that assess Fc-functional activities include measurements of antibody-dependent cellular cytotoxicity and complement-dependent cytotoxicity and are expected to corroborate results from in vitro target and Fc receptor binding assays. Differences in charge variants, mass, and absorbance are compared, as are degradation trends between the biosimilar and RP.

Appreciation for the high degree of sensitivity of the assays used in the CAA and their ability to predict clinical performance continues to grow.³⁰⁻³⁴ Although there have been general improvements in technology used in CAAs, it was the significant experience the FDA and other regulators gained in evaluating CAAs that supported confidence in streamlined clinical expectations. As described in the FDA's recently published draft guidance,¹⁴ "currently available analytical technologies can structurally characterize highly purified therapeutic proteins and model in vivo functional effects with a high degree of specificity and sensitivity using in vitro biological and biochemical assays." This means that molecular structure can be carefully examined with an understanding of how each structural element could relate to or affect a clinically relevant measure. For example, the glycosylation patterns could influence a wide range of clinical features,⁴⁶⁻⁴⁸ including half-life, immunogenicity, and efficacy by differences in Fc receptor binding and immune effector cell function. Sufficient scientific evidence has been accrued to help inform whether an observed analytical difference is of concern or has the potential to affect clinical performance.

Process-related impurities that may occur could derive from media components, residual solvents, host cell protein and DNA, microbiologic contaminants, material properties, and formulation (Figure 3). As with all biologics approved by the FDA, these are controlled during the manufacture of biosimilars and RPs alike. Both RP and biosimilars must meet the FDA's standards required for all biologics to ensure the drug product is consistently safe, pure, and potent. An assessment of toxicity without animal studies^{49,50} is performed comparing the quantitative and qualitative composition of each component of the biosimilar finished dosage form or drug product with the RP drug product. This includes potential differences in excipients: inactive substances in the media and potential extractables and leachables of the container closure systems. Tight control of these manufacturing aspects of biologics is essential to ensure product safety.

Synthesis: a CAA and a human PK study are generally sufficient to determine biosimilarity

The current state of biosimilar development is undergoing a Hegelian synthesis between the long-held thesis that CESs are useful for confirming the findings of the CAA and determining biosimilarity, and the antithesis counter approach where the CAA's ability to detect analytical differences between a proposed biosimilar and a RP is more useful for identifying potential differences in clinical performance than a CES (Figure 4). Recent research supports this new paradigm demonstrating that clinical data has a limited role in resolving uncertainty identified from the CAA for biosimilars.^{16-18,51-54} These reports include an analysis of 36 marketing applications by the European Medicines Agency (EMA) where "seemingly negative clinical data" were observed in the CES but the analytical/product quality assessments supported

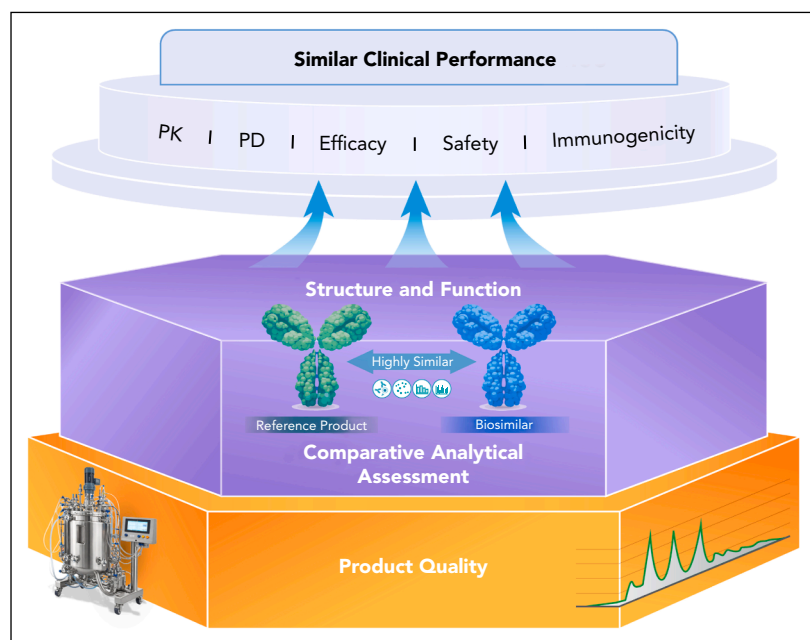


Figure 4. The foundation supporting biosimilarity. A synthesis of how the CAA coupled with product quality controls provides confidence that the biosimilar will have similar clinical outcomes as the RP with respect to PK, pharmacodynamics, efficacy, safety, and immunogenicity: the development of antidrug antibodies. PD, clinical study with pharmacodynamic end points; PK, human pharmacokinetic study.

approval. In each case, unexpected clinical results were attributed to imbalances in arms, immature secondary end point results, or other chance findings not related to the biosimilar.¹⁸ In a review of 7 adalimumab and 5 bevacizumab biosimilar candidates, CES data were found to have a limited role in addressing quality concerns in cases where values within the CAA for biosimilars were outside the acceptable RP similarity range. In these cases when an adequate justification could not be provided, the candidate biosimilar was not considered approvable.¹⁷

The CAA identifies differences in attributes between a proposed biosimilar and RP that could affect changes in PK, immunogenicity, efficacy, and safety, and is more sensitive and efficient than relying on results from a CES to evaluate the likelihood of differences.^{18,51} It is not surprising that findings from a CES alone have not been used as the basis for concluding 2 products are not biosimilar.^{16,55} When analytical differences between proposed biosimilars and their RP are observed, sufficient scientific evidence has been accrued to help understand the potential impact of those analytical differences on clinical performance.

The safety assessment for a new biologic is typically a descriptive review of results from ≥ 1 clinical trials to determine the safety profile of the new biologic. In contrast, the expected safety profile of the biosimilar is known from the labeling for the RP. A reasonably powered CES is unlikely to identify rare clinical events or differentiate occurrence rates between the proposed biosimilar and its RP. Rare safety events, should they occur, are expected to be the same for biosimilars and their RP and are expected to be captured in routine postmarketing safety reporting. Additionally, CESs are expensive and the potential expectation for and cost of a CES contribute to the risk of a "biosimilar void."⁵⁶ CESs also require enrolling patients who would otherwise be enrolled in trials for new treatments.

Outside the United States, patients are routinely switched from RP to biosimilar or from biosimilar to a second biosimilar without prescriber intervention without changes in efficacy or safety following the switch.²⁶ In the United States, the BPCIA created the interchangeable designation for biosimilars, which developers may choose to seek. An interchangeable biosimilar can be substituted for the RP at pharmacies without the involvement of the prescriber, subject to state laws. The FDA initially expected applicants to provide clinical switching studies (where patients undergo multiple switches between the RP and biosimilar) to support a demonstration of interchangeability; however, experience has shown that the risk in terms of safety or diminished efficacy is insignificant following single or multiple switches between a RP and biosimilar.^{25,57} Regardless of whether the developer seeks the interchangeable designation, all approved biosimilars are subject to the same standards and are as safe as the RP.

Regulatory health authorities, including the FDA, have issued draft guidance or reflection papers with opportunities for public comment that CESs are not generally necessary.^{14,20,52,58-63} Previously, the FDA and other global regulators generally recommended a CES for most biosimilar programs, including biosimilars for hematology indications. Although the high cost of development and the stagnation of

the biosimilar pipeline for many off-patent biologics⁶ are due to business decisions, "market failure" is not the driver for this change in thinking. Revised recommendations are instead driven by the evolution of scientific understanding and growing confidence in CAA data. The updated FDA guidance published in October 2025 describes recommendations for assessing the need for CESs and when a CES may not be necessary to support a demonstration of biosimilarity.¹⁴ It states that the streamlined approach applies when the RP and proposed biosimilar product are manufactured from clonal cell lines, are highly purified, and can be well characterized analytically; the relationship between quality attributes and clinical efficacy is generally understood for the RP, and these attributes can be evaluated by assays included in the CAA; and a PK similarity study is feasible and clinically relevant. Although the FDA, Medicines and Healthcare Products Regulatory Agency, EMA, and Health Canada have proposed similar approaches for when a CES is needed, as part of an International Council on Harmonisation Multidisciplinary guideline, regulatory agencies and other stakeholders are coalescing around a common approach for determining when a CES may be useful.^{16,17,52,53,61-64} Building consensus will harmonize development expectations across jurisdictions and support streamlined global biosimilar development.

Future directions

Biosimilars used to treat hematologic conditions are now widely available and generally accepted by hematologists. Since the passage of the BPCIA, biosimilars have been approved to treat patients with a range of conditions with approvals being based primarily on a combination of a CAA plus clinical PK-PD studies or CAA plus clinical PK and CES. Given the increased awareness that CESs are not likely to be informative if the CAA and PK study results support biosimilarity and efforts by global regulatory authorities to clarify when a CES may be useful for a biosimilar program, CESs are expected to be conducted much less frequently.¹⁴ Practical implications of this change may be more rapid development and approval of biosimilars with lower associated costs. Although clinical trials, through their key role in the approval process for originator biologics, may continue to be viewed as the standard, the FDA through ongoing stakeholder engagement and educational resources will address issues related to prescriber hesitancy. Regardless of whether a development program includes a CES, high standards for all biosimilars are maintained. Patients and health care providers can be confident that biosimilars are as safe and effective as their RP, offering the same risks and benefits.

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Footnotes

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